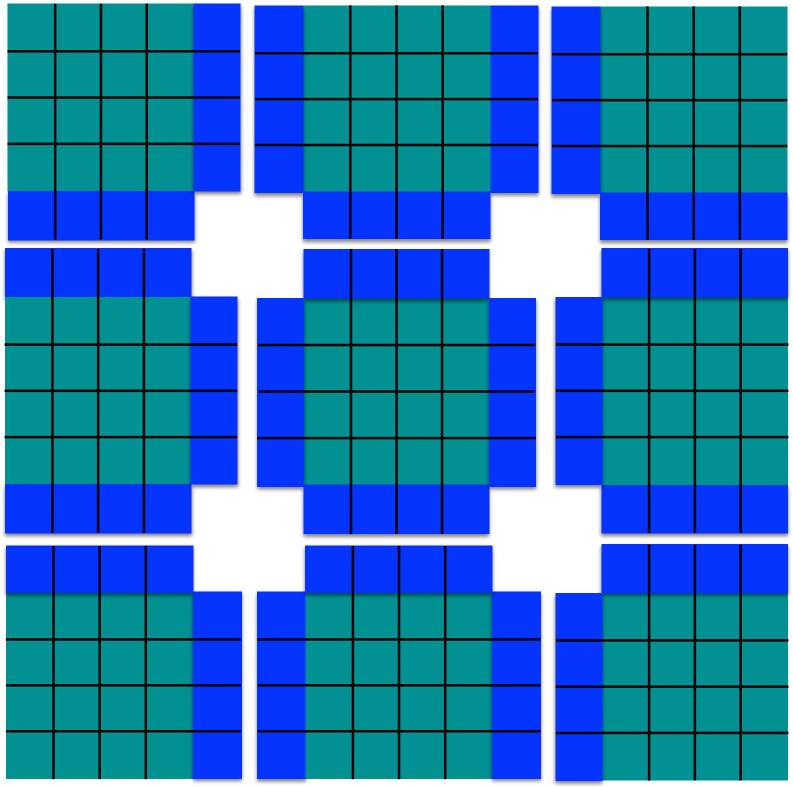
Domain decomposition overhead

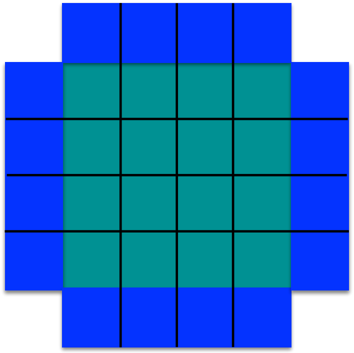
What is the communication overhead when we decompose a 2-dimensional domain?



3x3 decomposition of 12x12 2D domain

Blue regions are haloes

Consider an NxN sub-domain with 4 neighbours:



Computation: $N \times N = N^2$
Communication: $4 \times N \times 1 = 4N$

Bandwidth limited

Ratio of communication to computation:

$$R_{2D} = \frac{4N}{N^2} = \frac{4}{N} \Rightarrow R_{2D} \propto \frac{1}{N}$$

3-dimensional domain

- For a 3-dimensional sub-domain of size NxNxN
- Computation: N^3
- Communication: $6N^2$ (cube has 6 faces)
- Ratio of communication to computation:

$$R_{3D} = \frac{6N^2}{N^3} = \frac{6}{N} \Rightarrow R_{3D} \propto \frac{1}{N}$$

Domain decomposition overhead

- In both the 2D and 3D cases the ratio of number of halo points to number of points in the sub-domain scales as $1/N$

- In strong scaling N decreases as the domain is decomposed further in order to use more processors

- The ratio of communication to computation increases as the sub-domain size decreases which adversely impacts strong scaling

- Storing halos also requires extra memory. We can reduce the memory required to store haloes if we combine OpenMP and MPI

Super-linear speed-up

- Strong scaling a domain decomposed calculation reduces the domain size per processor
- Sometimes an important data structure becomes small enough to fit into memory cache
- This can result in a sudden performance increase and a better than ideal (super-linear) speed-up

Unit Summary

- In a domain decomposed calculation the ratio of halo points to domain points scales as $1/N$

- As we strong scale a domain decomposed calculation to increasing numbers of processors the ratio of communication to computation increases

- Strong scaling a domain decomposed calculation decreases the memory footprint per processor. We can see a super-linear speed up if a significant data structure fits into cache